



NPTEL Course on “Computer System Optimization for ML”

Lecture #9



Agenda

- **Recap**
- **More CNNs**

NPTREL

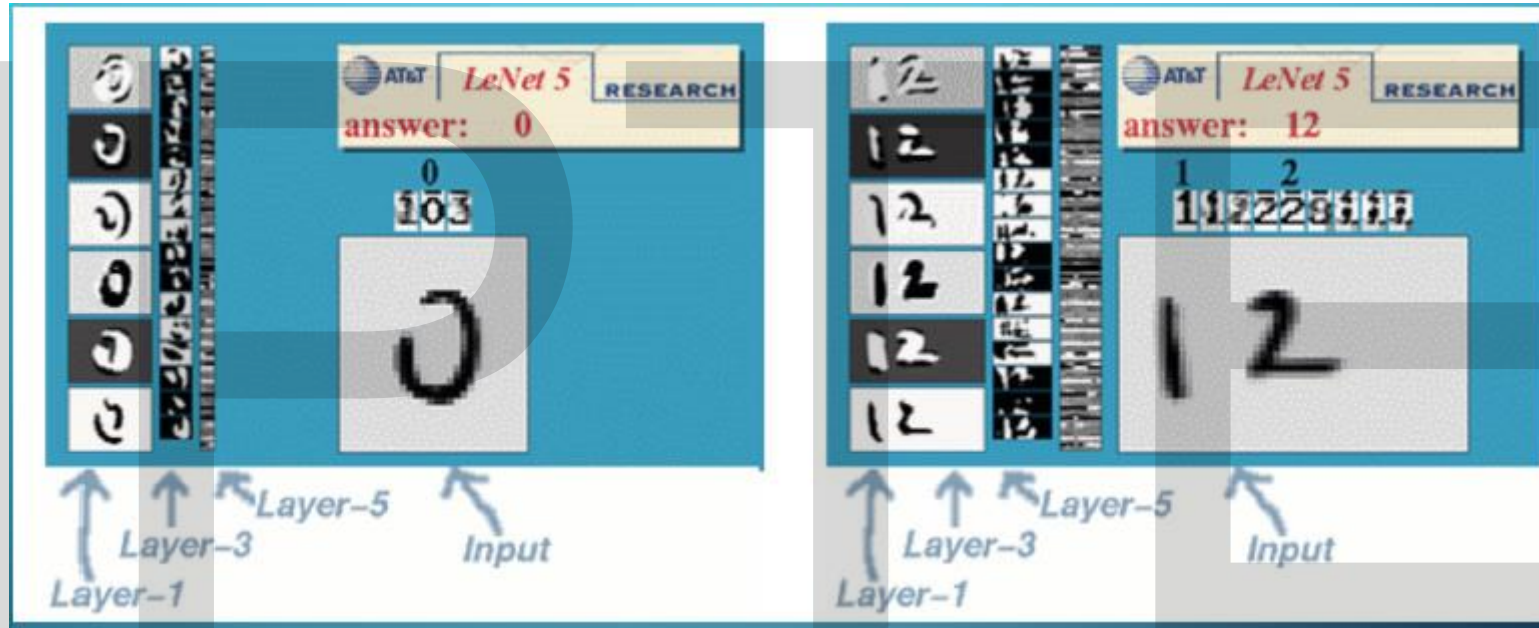
Commonly used Dataset

■ MNIST

- Digit Classification
- 28×28 pixels (B&W)
- 10 Classes
- 60,000 training
- 10,000 testing



LeNet-5



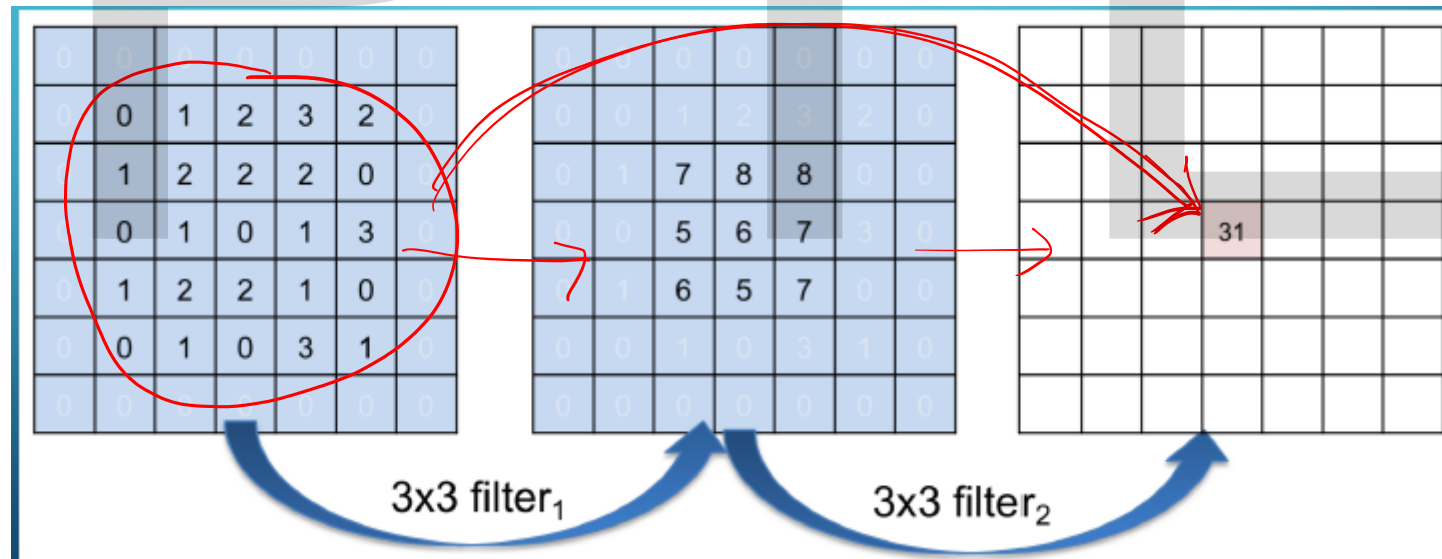
ImageNet

- **Image classification**
 - 256×256
 - Colour images
 - 1000 classes
 - 1.3 M training
 - 100,000 testing
- **For ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**
 - Accuracy of classification task reported based on top-1/top-5 error



Stacked Filters

- Deeper networks means more weights
- Use stack of smaller filters (3×3) to cover the same receptive field with fewer filter weights
- Non-linear activations inserted between each filter
 - 5×5 filter (25 weights) → two 3×3 filters (18 weights)



VGG-16 (Simonyan et al., ICLR 2015)

- Conv layers: 13
- FC layers: 3
- Weights: 138M
- MACs: 15.5G
- There is a 19-layer version too (VGG-19)

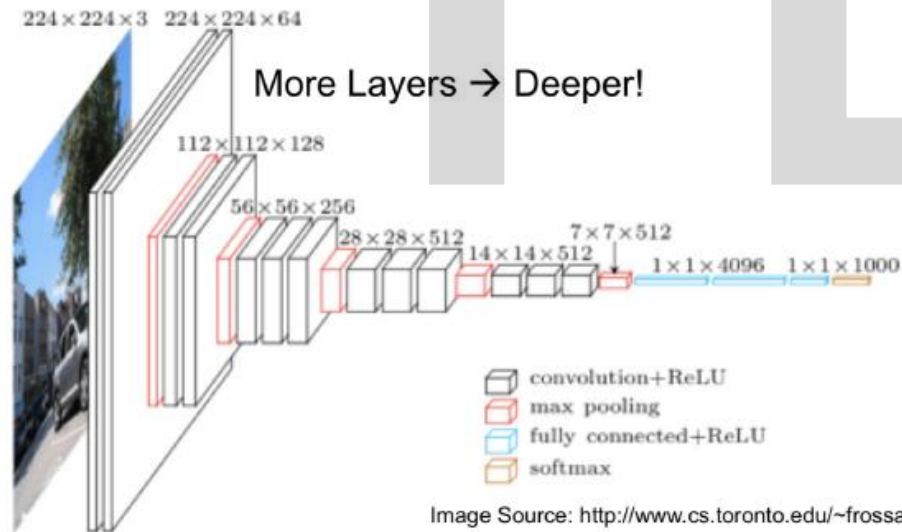


Image Source: <http://www.cs.toronto.edu/~frossard/post/vgg16/>

We are not Done Yet!



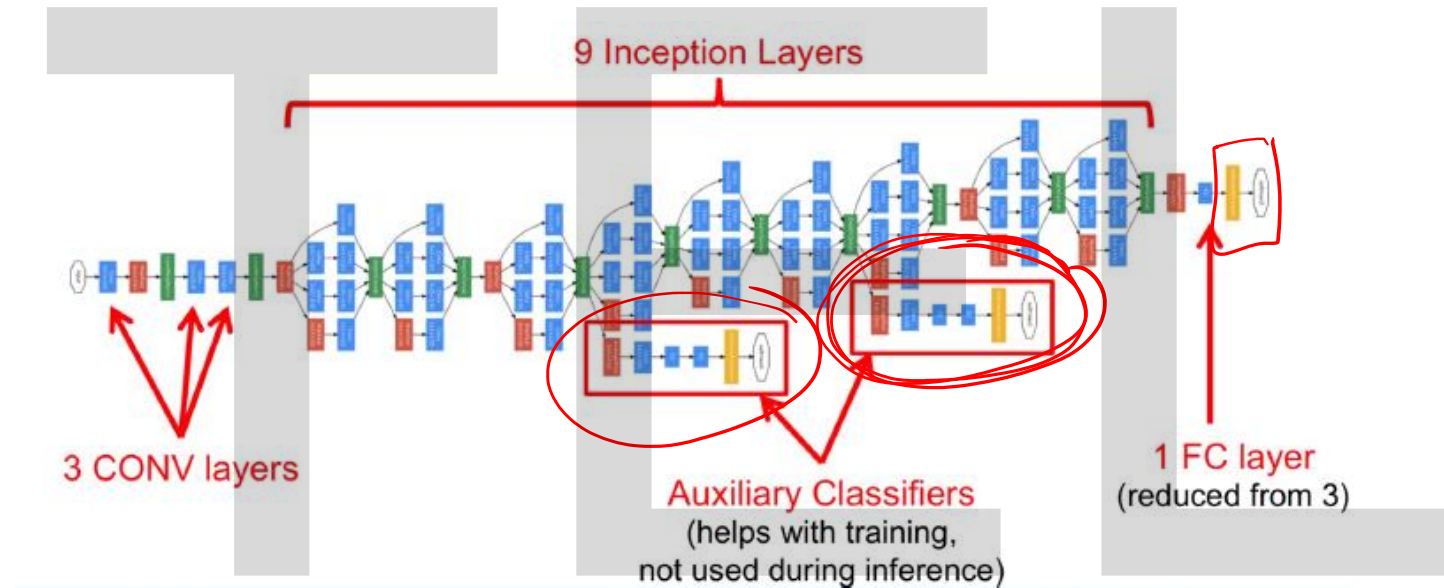
Agenda

- Recap
- **More CNNs**

NPTREL

GoogleNet/Inception (V1)

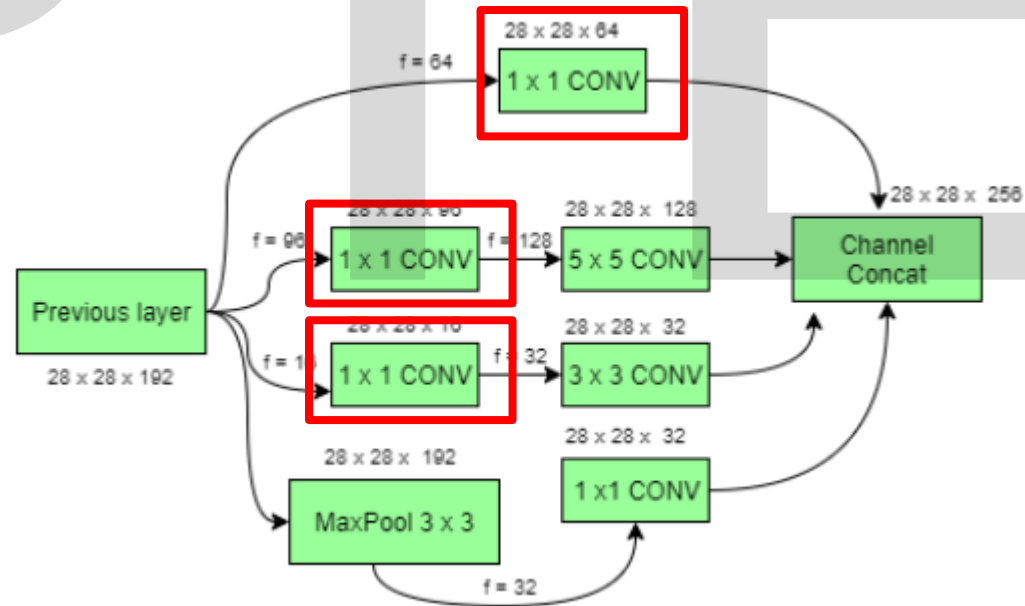
- Conv Layers
 - 57 (total), 21 (depth) (?)
- Fully Connected Layers: 1
- Weights: 7M
- MACs: 1.43G
- V2, V3 and V4 are ILSVRC 14 winner



Deep into Inception

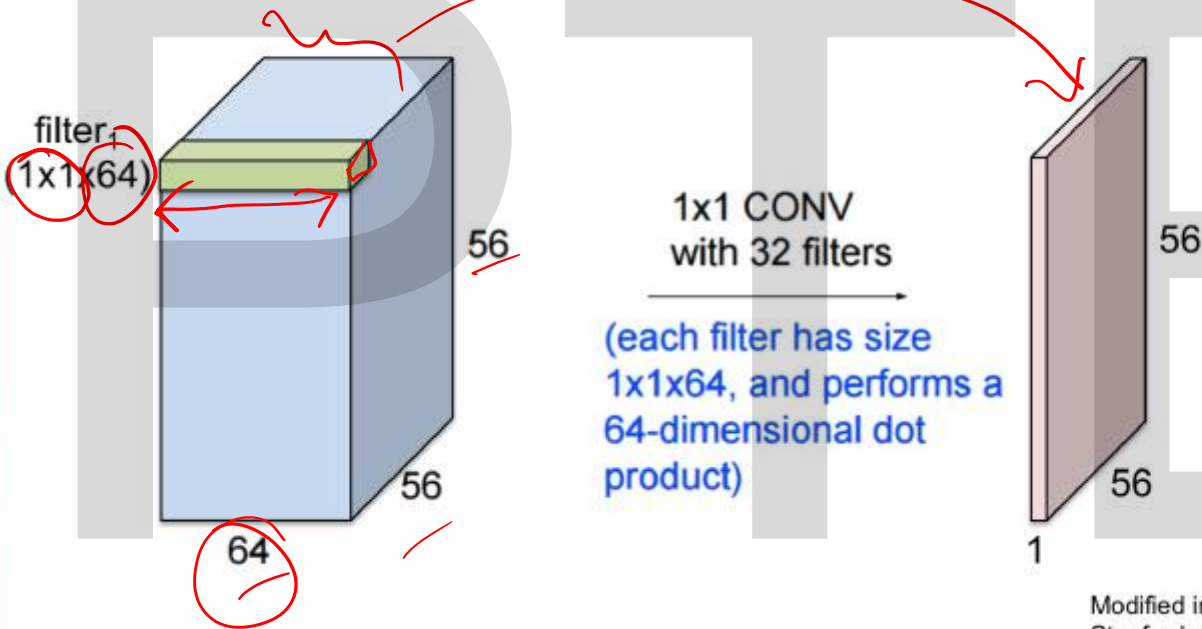


- 1×1 Conv reduce the channel depth
- Auxiliary loss: ~~loss~~^{loss} in the inception module
 - Total loss = $\alpha \times \text{real loss} + (1 - \alpha) \times \text{auxiliary loss}$
 - Solves vanishing gradient problem



1x1 Convolution

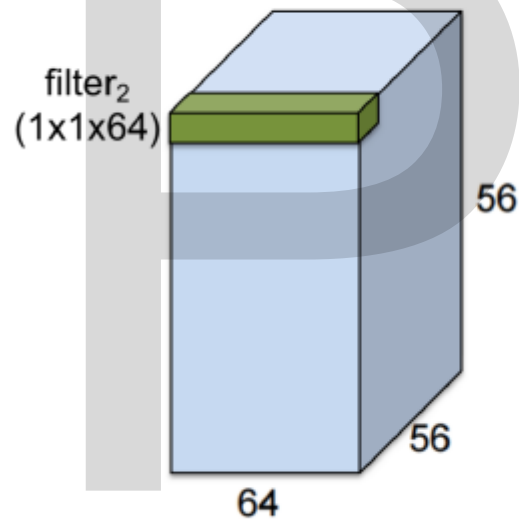
Use **1x1 filter** to capture cross-channel correlation, but no spatial correlation.
Can be used to reduce the number of channels in next layer (**bottleneck**)



Modified image from source:
Stanford cs231n

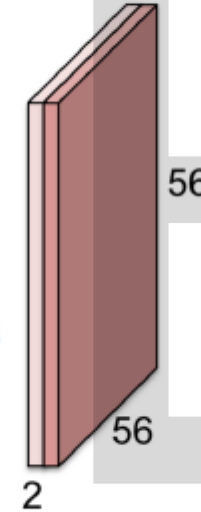
1×1 Convolution

Use **1x1 filter** to capture cross-channel correlation, but no spatial correlation.
Can be used to reduce the number of channels in next layer (**bottleneck**)



1x1 CONV
with 32 filters

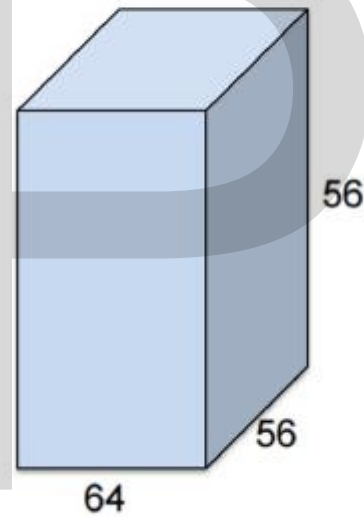
(each filter has size
1x1x64, and performs a
64-dimensional dot
product)



Modified image from source:
Stanford cs231n

1×1 Convolution

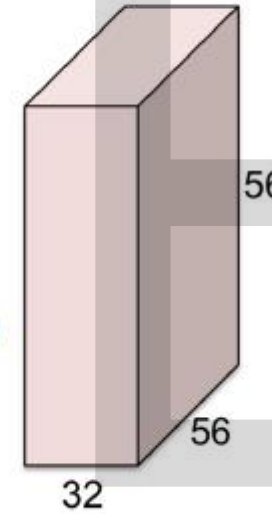
Use **1x1 filter** to capture cross-channel correlation, but no spatial correlation.
Can be used to reduce the number of channels in next layer (**bottleneck**)



1x1 CONV
with 32 filters

→

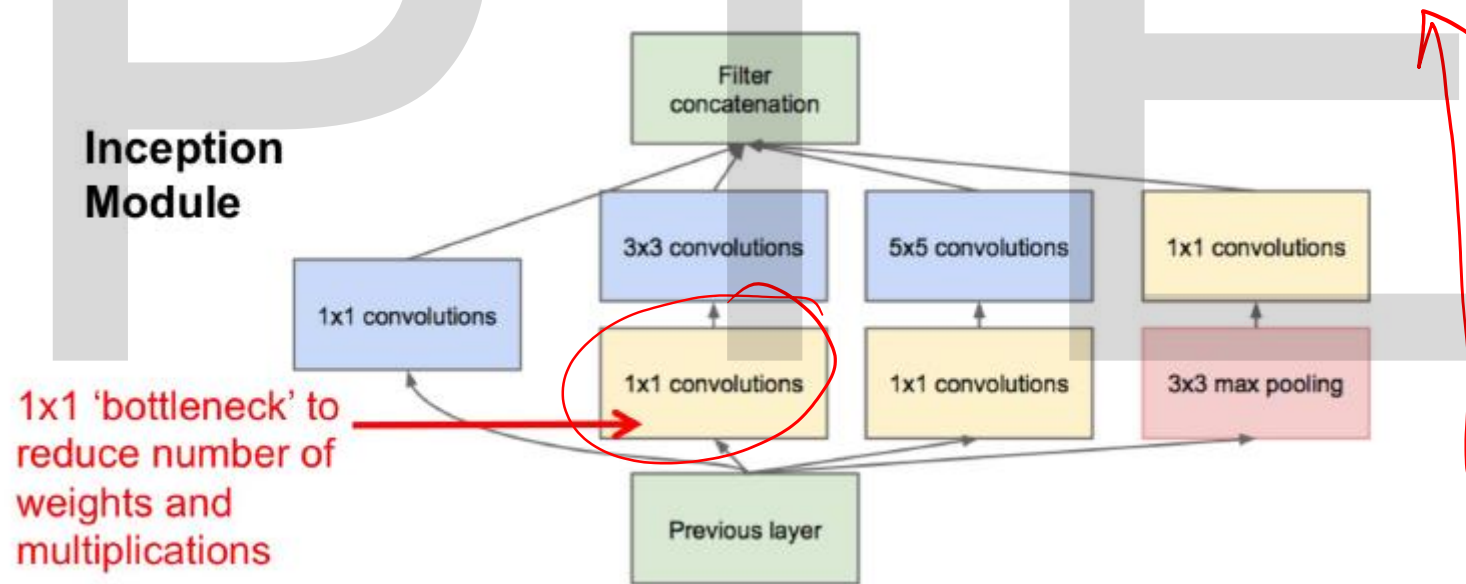
(each filter has size
1x1x64, and performs a
64-dimensional dot
product)



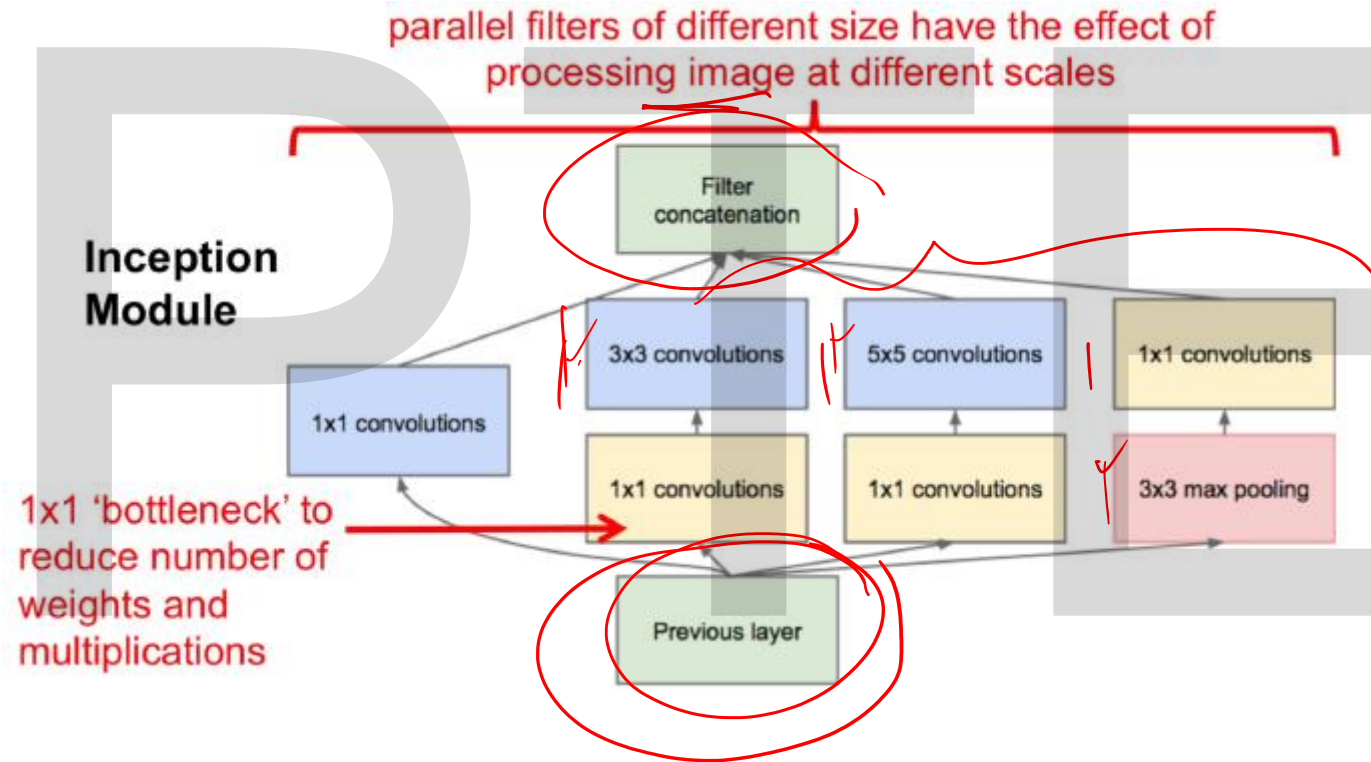
Modified image from source:
Stanford cs231n

Inception V1

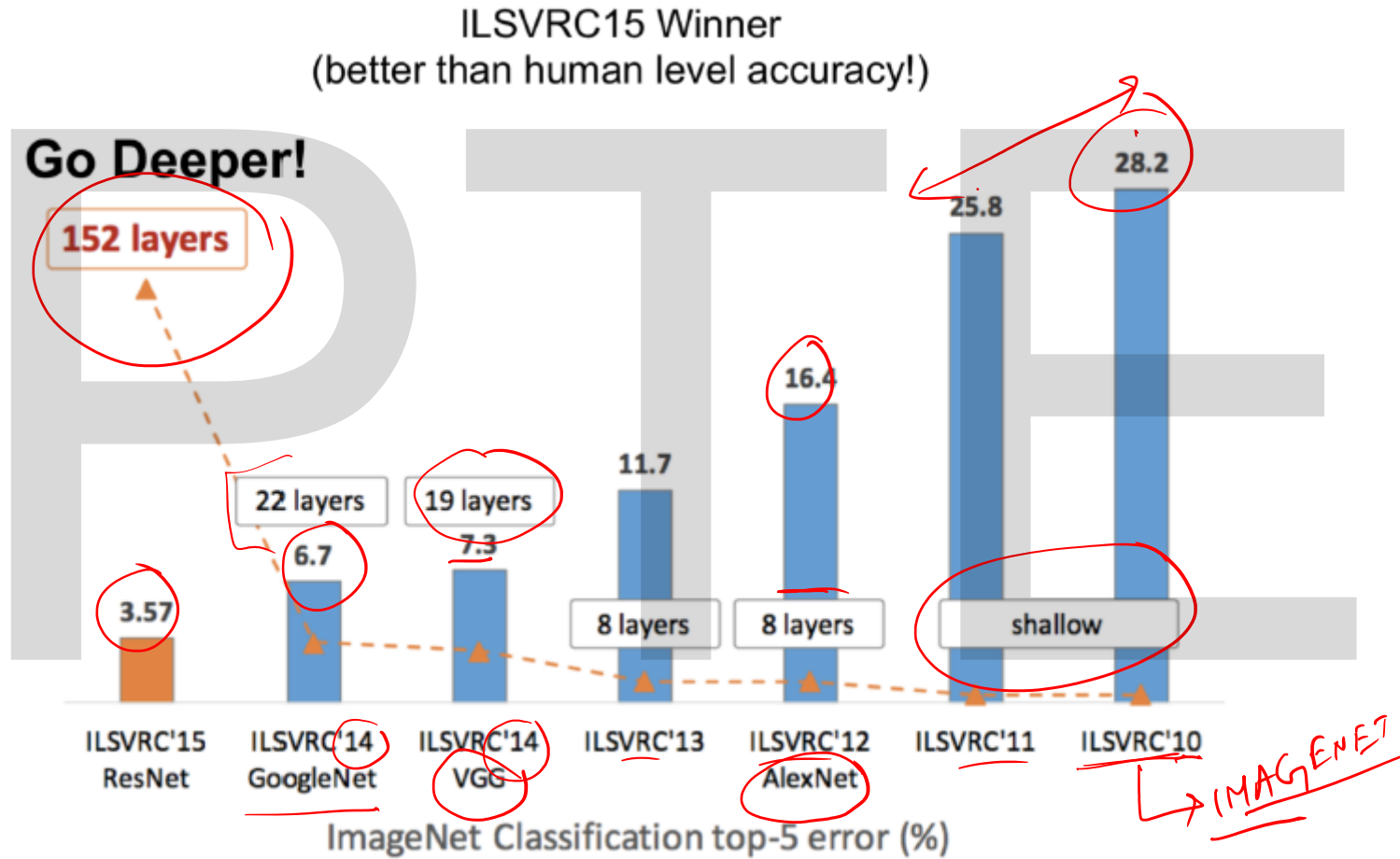
- Apply 1×1 before 'large' convolution filters
- Reduce weights such that the entire DNN can be trained on one GPU
- Number of multiplications reduced from 854M $\rightarrow$ 358M



Deep into Inception

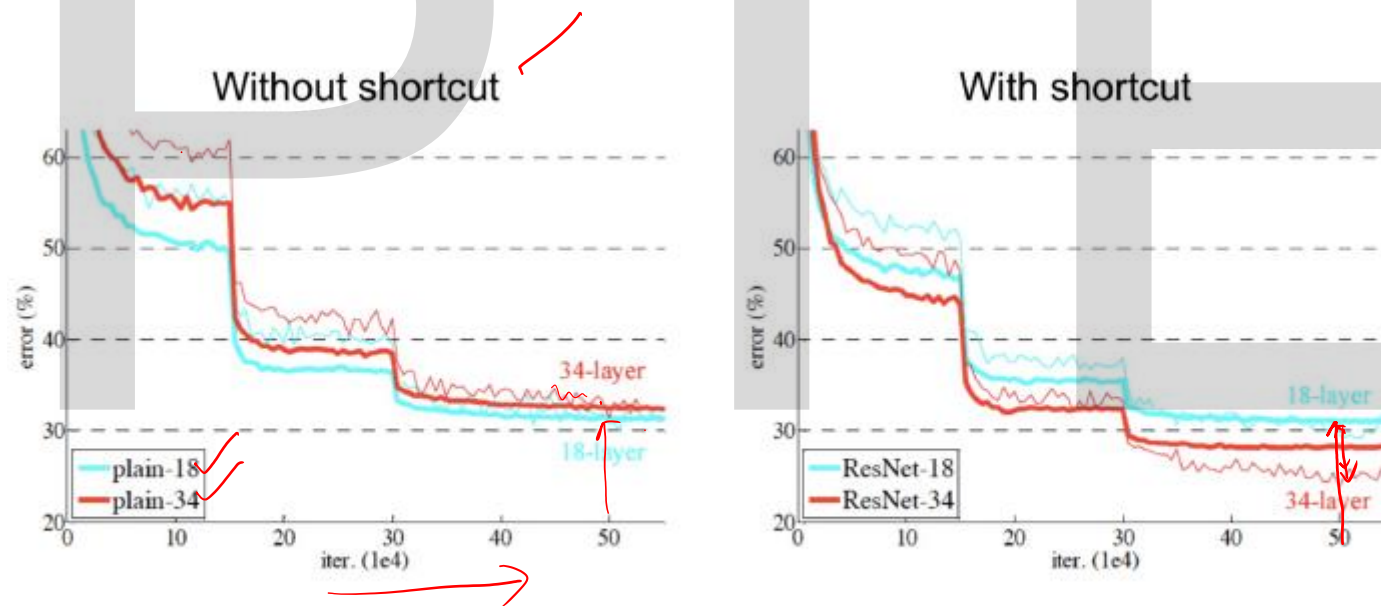


RESNET (He et al., CVPR, 2016)



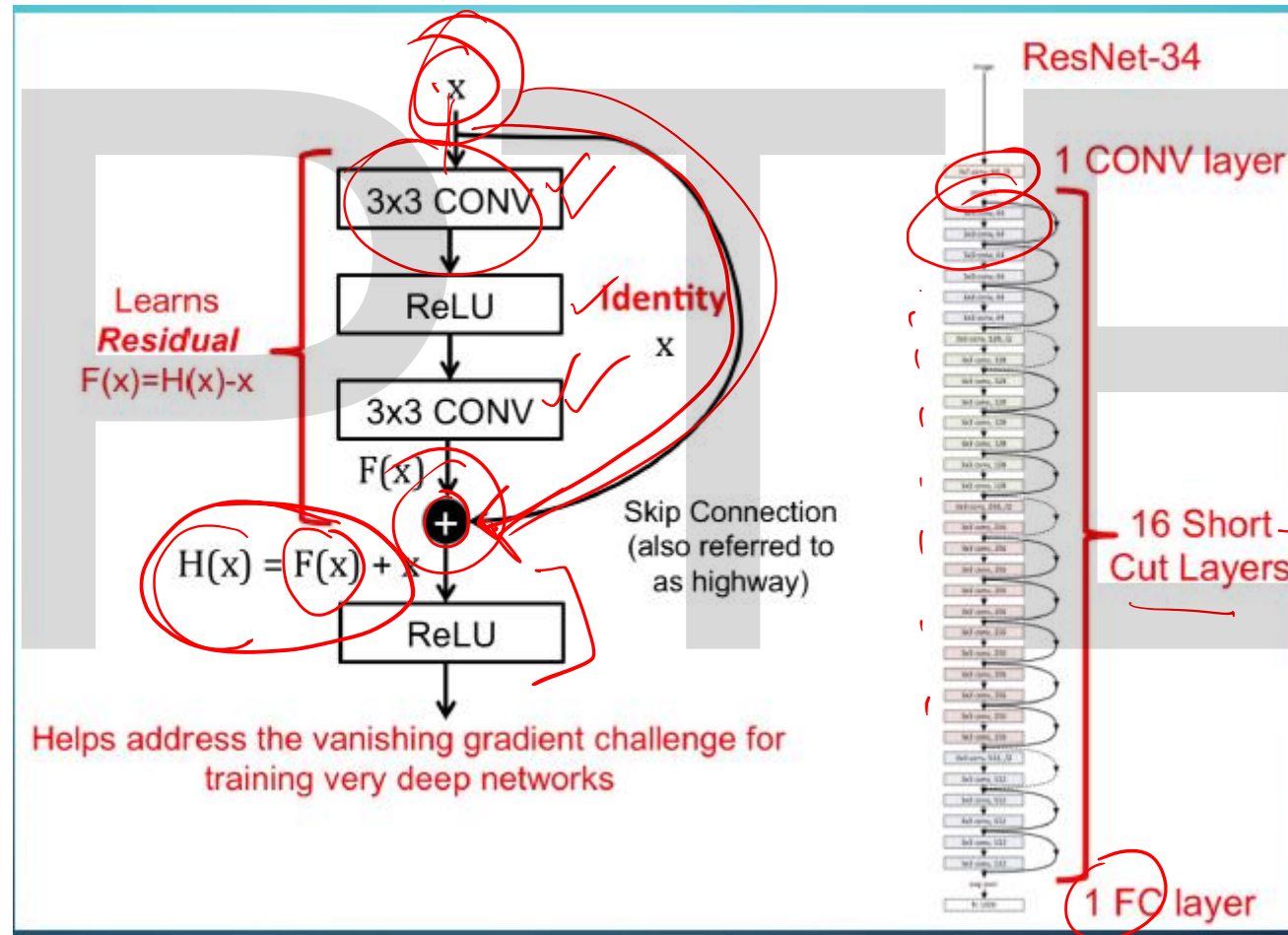
RESNET

- Training and validation error increases with more layers
 - Vanishing gradient
- Introduce 'short cut modules' to address the problem



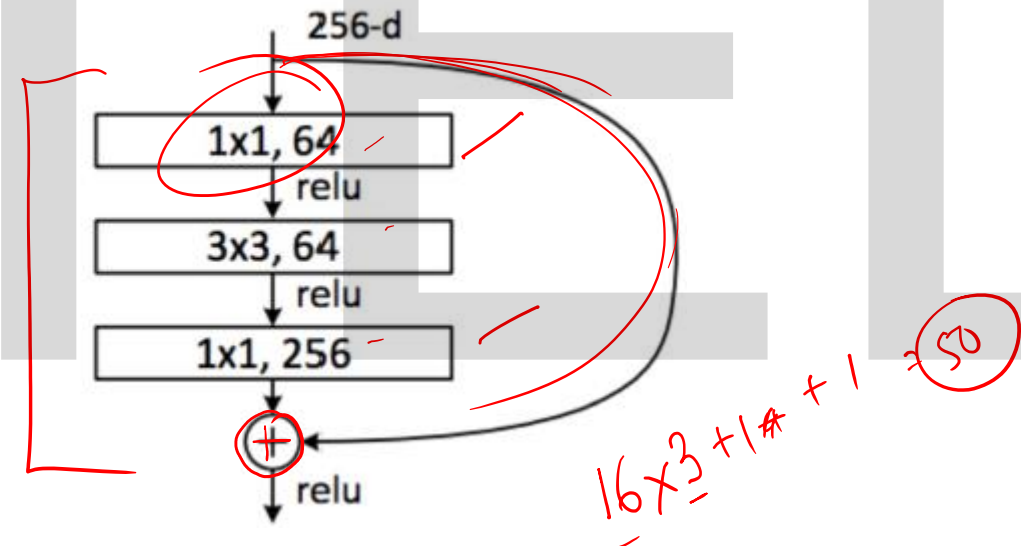
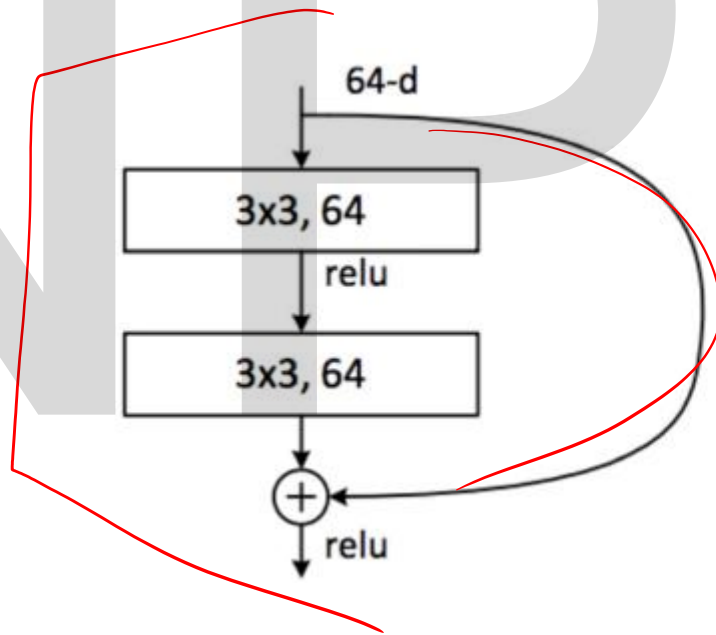
Thin curves denote training error, and bold curves denote validation error.

RESENET: Short Cut Module



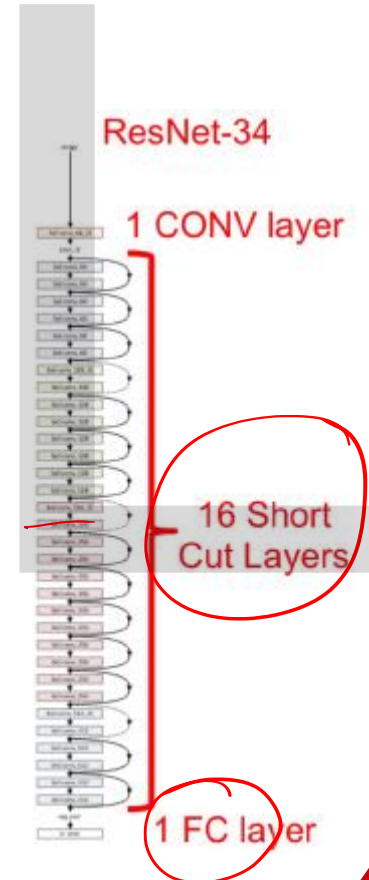
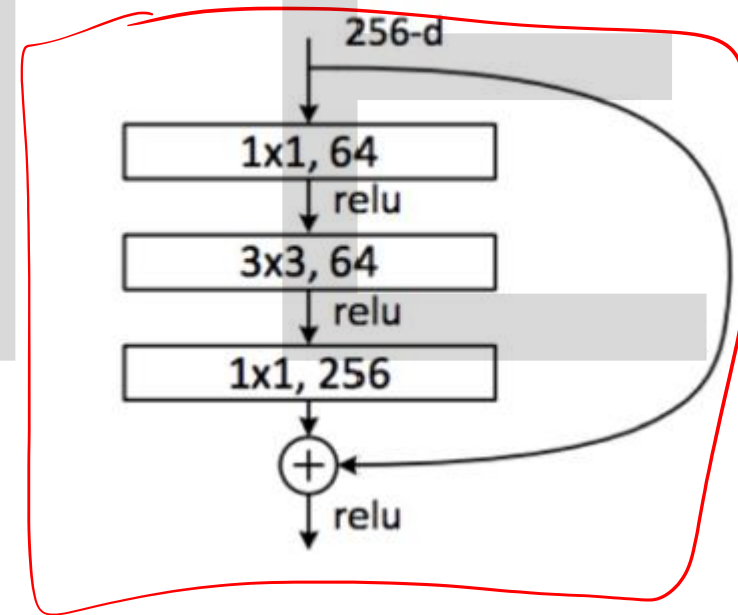
RESNET: Use of 1x1 Filters

- Apply 1x1 filters to reduce computation and size
- Makes the network deeper (ResNet 34 → ResNet 50)



RESNET-50

- Conv layers: 49
- FC layers: 1
- Weights: 25.5M
- MACs: 3.9G
- There exist 34, 152, 1202 versions also





THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)

